Solutions

Maths

Q1. Let $f(x)$ be a function defined by $f(x) = \int \frac{x^2 e^x}{(x+2)^2} dx$. If the curve $y = f(x)$ touches the straight line $y = \frac{e^2}{4}x + 1$ at $x = 2$, then the value of $f(0)$ is:

(A) $\frac{e^2}{4}$ (B) $\frac{e^2}{2}$ (C) e^2 (D) $e^2 - 1$

Ans. B

Sol. $f(x) = \int e^x \frac{x^2 - 4 + 4}{(x+2)^2} dx$

$= \int e^x \left(\frac{x-2}{x+2} + \frac{4}{(x+2)^2} \right) dx$

Since the integral is of the standard form $\int e^x (h(x) + h'(x)) dx = e^x h(x) + C$, we have:

$f(x) = e^x \left(\frac{x-2}{x+2} \right) + C$

Given that the curve $y = f(x)$ touches the line $y = \frac{e^2}{4}x + 1$ at $x = 2$, their y-values must be equal at the point of tangency:

$f(2) = \frac{e^2}{4}(2) + 1$

$f(2) = e^2 \left(\frac{2-2}{2+2} \right) + C = C$

$C = \frac{e^2}{2} + 1$

Now, we evaluate $f(0)$:

$f(0) = e^0 \left(\frac{0-2}{0+2} \right) + C = -1 + C$

$f(0) = -1 + \left(\frac{e^2}{2} + 1 \right) = \frac{e^2}{2}$

Q2. The value of the limit

$\lim_{x \to 0} \left(\frac{e^x \ln(1+x) + e^{-x} \ln(1-x)}{x^2} \right)^{\frac{1}{1-\cos(x^2)}}$

is equal to:

(A) $e^{-\frac{7}{36}}$ (B) $e^{-\frac{7}{72}}$ (C) $e^{-\frac{5}{36}}$ (D) $e^{-\frac{1}{12}}$

Ans. A

Sol. Let the given limit be L. The limit is of the indeterminate form 1^{∞}.

$$\ln L = \lim_{x\to 0} \frac{\frac{e^x \ln(1+x)+e^{-x}\ln(1-x)}{x^2} - 1}{1-\cos(x^2)}$$

$$\ln L = \lim_{x\to 0} \frac{e^x \ln(1+x) + e^{-x}\ln(1-x) - x^2}{x^2(1-\cos(x^2))}$$

Using the standard Taylor series expansions about $x = 0$:

$$e^x \ln(1+x) = x + \frac{1}{2}x^2 + \frac{1}{3}x^3 + 0\cdot x^4 + \frac{3}{40}x^5 - \frac{7}{144}x^6 + \dots$$

Replacing x with $-x$, we get:

$$e^{-x}\ln(1-x) = -x + \frac{1}{2}x^2 - \frac{1}{3}x^3 + 0\cdot x^4 - \frac{3}{40}x^5 - \frac{7}{144}x^6 + \dots$$

$$e^x \ln(1+x) + e^{-x}\ln(1-x) = x^2 - \frac{7}{72}x^6 + O(x^8)$$

$$\ln L = \lim_{x\to 0} \frac{\left(x^2 - \frac{7}{72}x^6 + O(x^8)\right) - x^2}{x^2\left(\frac{x^4}{2} - \frac{x^8}{24} + \dots\right)} \Rightarrow \ln L = \lim_{x\to 0} \frac{-\frac{7}{72}x^6 + O(x^8)}{\frac{1}{2}x^6 - \frac{1}{24}x^{10} + \dots}$$

$$\ln L = \frac{-\frac{7}{72}}{\frac{1}{2}} = -\frac{7}{36} \Rightarrow \text{Thus, } L = e^{-\frac{7}{36}}.$$

Q3. Let a triangle have vertices $A(1,1)$, $B(5,1)$, and $C(1,4)$. A family of straight lines is represented by the equation: $(6\lambda+3)x + (3\lambda-6)y - (20\lambda-5) = 0$ where $\lambda \in \mathbb{R}$ is a parameter. If the maximum distance of the orthocenter of $\triangle ABC$ from any line of this family is d, then the value of $3d$ is:

(A) 4 (B) 5 (C) 6 (D) 7

Ans. B

Sol. The side AB lies on the line $y = 1$ (horizontal), and the side AC lies on the line $x = 1$ (vertical). Since $AB \perp AC$, $\triangle ABC$ is a right-angled triangle at $A(1,1)$.

In a right-angled triangle, the orthocenter H is the vertex containing the right angle. Thus, $H \equiv (1,1)$.

Rearranging the family of lines in the standard $L_1 + \lambda L_2 = 0$ form:

$(3x - 6y + 5) + \lambda(6x + 3y - 20) = 0$

Thus, the family of lines always passes through the fixed point $G\left(\frac{7}{3}, 2\right)$, which matches the centroid of $\triangle ABC$ since:

$$\text{Centroid} = \left(\frac{1+5+1}{3}, \frac{1+1+4}{3}\right)$$

$$= \left(\frac{7}{3}, 2\right)$$

The maximum distance d from a fixed point H to a variable line passing through another fixed point G is equal to the length of the segment HG itself (occurring when the line is perpendicular to HG).

$$d = HG = \sqrt{\left(\frac{7}{3} - 1\right)^2 + (2-1)^2}$$

$$d = \sqrt{\left(\frac{4}{3}\right)^2 + 1^2}$$
$$= \sqrt{\frac{16}{9} + 1}$$
$$= \sqrt{\frac{25}{9}}$$
$$= \frac{5}{3}$$

Therefore, the value of $3d$ is:

$3d = 3 \times \frac{5}{3} = 5$

Q4. Let C be the curve defined by $y = \frac{\ln x}{x}$ for $x > 0$. Let S be the set of all real numbers a such that exactly two distinct tangents can be drawn from the point $(a, 0)$ to the curve C. Then the set S is equal to

(A) $(0,1) \cup (4e^{3/2}, \infty)$ (B) $(0,1] \cup [4e^{3/2}, \infty)$ (C) $(-\infty, 0) \cup (1, 4e^{3/2})$

(D) $(0, e) \cup (e^{3/2}, \infty)$

Ans. A

Sol. The given curve is $y = f(x) = \frac{\ln x}{x}$ for $x > 0$.

$f'(x) = \frac{1 - \ln x}{x^2}$

Let $P\left(t, \frac{\ln t}{t}\right)$ be the point of tangency on C (where $t > 0$). The equation of the tangent line at P is:

$y - \frac{\ln t}{t} = \frac{1 - \ln t}{t^2}(x - t)$ $y = \frac{1 - \ln t}{t^2}x + \frac{2\ln t - 1}{t}$

Since the tangent passes through the point $(a, 0)$, substituting $x = a$ and $y = 0$ gives:

$0 = \frac{1 - \ln t}{t^2}a + \frac{2\ln t - 1}{t} \implies a(1 - \ln t) = t(1 - 2\ln t)$

For $t \neq e$, we can express a as a function of t: $a = g(t) = \frac{t(2\ell n\, t - 1)}{\ell n\, t - 1}$

The number of distinct tangents from $(a, 0)$ equals the number of real roots of $g(t) = a$. Let's analyze $g(t)$ by differentiating it:

$$g'(t) = \frac{(2\ln t + 1)(\ln t - 1) - (2\ln t - 1)}{(\ln t - 1)^2}$$
$$= \frac{\ln t(2\ln t - 3)}{(\ln t - 1)^2}$$

Setting $g'(t) = 0$ gives the critical points:

$\ln t = 0 \implies t = 1$

$2\ln t - 3 = 0 \implies t = e^{3/2}$

$g(1) = \frac{1(-1)}{-1} = 1$ $g\left(e^{3/2}\right) = \frac{e^{3/2}(3-1)}{1.5 - 1} = 4e^{3/2}$

Analyzing the behavior of $g(t)$ across its domain $(0,\infty)\setminus\{e\}$:

- For $t\in(0,1)$, $g'(t)>0 \implies g(t)$ strictly increases from 0 to 1.
- For $t\in(1,e)$, $g'(t)<0 \implies g(t)$ strictly decreases from 1 to $-\infty$.
- For $t\in(e,e^{3/2})$, $g'(t)<0 \implies g(t)$ strictly decreases from $+\infty$ to $4e^{3/2}$.
- For $t\in(e^{3/2},\infty)$, $g'(t)>0 \implies g(t)$ strictly increases from $4e^{3/2}$ to $+\infty$.

By plotting $a=g(t)$, the number of intersections with the horizontal line $y=a$ is exactly two if and only if $a\in(0,1)$ or $a\in(4e^{3/2},\infty)$.

Thus, $S=(0,1)\cup(4e^{3/2},\infty)$.

Q5. Let C_1 be a curve passing through the point $(1,1)$ and satisfying the differential equation $2y\dfrac{dy}{dx}=3x^2$ for all $x>0, y>0$. Let C_2 be a member of the family of ellipses given by $3x^2+4y^2=\lambda$, where $\lambda>0$ is a real parameter. If C_1 and C_2 intersect at a point $P(x_0,y_0)$ in the first quadrant, then which of the following statement(s) is (are) **TRUE**?

(A) The product of the slopes of the tangents to C_1 and C_2 at P is independent of λ and is equal to $-\dfrac{9}{8}$.

(B) If $\lambda=7$, the acute angle θ of intersection between C_1 and C_2 at P satisfies $\tan\theta=18$.

(C) The area of the region bounded by the curve C_1, the x-axis, and the vertical line passing through P when $\lambda=7$ is equal to $\dfrac{2}{5}$.

(D) There exists a unique value of $\lambda>0$ for which the curves C_1 and C_2 intersect orthogonally at P.

Ans. A, B, C

Sol. To find the equation of the curve C_1, we integrate its differential equation:

$$2y\frac{dy}{dx}=3x^2 \implies \int 2y\,dy=\int 3x^2\,dx \implies y^2=x^3+c$$

Since C_1 passes through $(1,1)$, we have $1=1+c \implies c=0$. Thus, C_1 is given by $y^2=x^3$.

Let $P(x_0,y_0)$ be the point of intersection in the first quadrant. Then $y_0^2=x_0^3$.

Differentiating the equation of C_1 gives the slope m_1 at P:

$$2y\frac{dy}{dx}=3x^2 \implies m_1=\frac{3x_0^2}{2y_0}$$

Differentiating the equation of C_2 ($3x^2+4y^2=\lambda$) gives the slope m_2 at P:

$$6x+8y\frac{dy}{dx}=0 \implies m_2=-\frac{3x_0}{4y_0}$$

The product of the slopes at P is:

$$\begin{aligned} m_1m_2 &= \left(\frac{3x_0^2}{2y_0}\right)\left(-\frac{3x_0}{4y_0}\right) \\ &= -\frac{9x_0^3}{8y_0^2} \end{aligned}$$

Since $y_0^2=x_0^3$, this simplifies to:

$m_1 m_2 = -\dfrac{9x_0^3}{8x_0^3} = -\dfrac{9}{8}$

This value is a constant and completely independent of λ. Hence, statement (A) is TRUE.

For $\lambda = 7$, the intersection point satisfies:

$3x_0^2 + 4x_0^3 = 7 \implies (x_0 - 1)(4x_0^2 + 7x_0 + 7) = 0$

The unique positive real root is $x_0 = 1$, which gives $y_0 = 1$. Thus, P is $(1, 1)$.

At $(1, 1)$, the slopes are $m_1 = \dfrac{3}{2}$ and $m_2 = -\dfrac{3}{4}$.

$$\begin{aligned}\tan\theta &= \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right| \\ &= \left|\frac{\frac{3}{2} - \left(-\frac{3}{4}\right)}{1 - \frac{9}{8}}\right| \\ &= \left|\frac{\frac{9}{4}}{-\frac{1}{8}}\right| \\ &= 18\end{aligned}$$

Hence, statement (B) is TRUE.

When $\lambda = 7$, P is $(1, 1)$. The area bounded by C_1 ($y = x^{3/2}$ for $y > 0$), the x-axis, and $x = 1$ is:

$$\begin{aligned}\text{Area} &= \int_0^1 x^{3/2}\,dx \\ &= \left[\frac{2}{5}x^{5/2}\right]_0^1 \\ &= \frac{2}{5}\end{aligned}$$

Hence, statement (C) is TRUE.

For orthogonal intersection, we must have $m_1 m_2 = -1$. However, we showed that $m_1 m_2 = -\dfrac{9}{8}$ identically at all intersection points. Thus, the curves can never intersect orthogonally for any $\lambda > 0$. Hence, statement (D) is FALSE.

Q6. Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined by: $f(x) = \min\left\{\max\left\{x + 1, |x^3 - x|\right\}, \max\left\{2 - x^2, \dfrac{2}{1 + |x|}\right\}\right\}$ Selecting $\phi = \dfrac{1 + \sqrt{5}}{2}$, which of the following statements is (are) **TRUE**?

(A) f is continuous on $\mathbb{R}$.

(B) The number of points in $\mathbb{R}$ where f is not differentiable is exactly 5.

(C) The maximum value of f on $\mathbb{R}$ is $\dfrac{\sqrt{5} - 1}{2}$.

(D) f is strictly decreasing on the interval $(1, \infty)$.

Ans. A, B, D

Sol. To analyze $f(x)$, let $A(x) = \max\left\{x + 1, |x^3 - x|\right\}$ and $B(x) = \max\left\{2 - x^2, \dfrac{2}{1 + |x|}\right\}$.

$$A(x) = \begin{cases} x - x^3 & \text{if } x < -1 \\ x^3 - x & \text{if } -1 \le x < 1 - \phi \\ x + 1 & \text{if } 1 - \phi \le x \le \phi \\ x^3 - x & \text{if } x > \phi \end{cases}$$

where $1-\phi=\dfrac{1-\sqrt{5}}{2}$.

$$B(x)=\begin{cases}\frac{2}{1-x} & \text{if } x<-1\\ 2-x^2 & \text{if } -1\le x\le 1\\ \frac{2}{1+x} & \text{if } x>1\end{cases}$$

Comparing $A(x)$ and $B(x)$ across different intervals:

1. For $x<-1$, $x-x^3=\dfrac{2}{1-x} \implies x^4-x^3-x^2+x-2=0$, which yields exactly one real root $x_0\in(-2,-1)$.
2. For $-1\le x<1-\phi$, $A(x)<B(x)$ holds throughout.
3. For $1-\phi\le x\le 1$, the intersection of $x+1$ and $2-x^2$ occurs at $x=\phi-1$.
4. For $x>1$, $B(x)<A(x)$ always holds.

$$f(x)=\begin{cases}\frac{2}{1-x} & \text{if } x<x_0\\ x-x^3 & \text{if } x_0\le x<-1\\ x^3-x & \text{if } -1\le x<1-\phi\\ x+1 & \text{if } 1-\phi\le x<\phi-1\\ 2-x^2 & \text{if } \phi-1\le x\le 1\\ \frac{2}{1+x} & \text{if } x>1\end{cases}$$

- Continuity: As the minimum of two continuous functions, $f(x)$ is continuous everywhere on $\mathbb{R}$. Thus, (A) is TRUE.
- Differentiability: Checking left and right derivatives at the transition boundaries $\{x_0,-1,1-\phi,\phi-1,1\}$ reveals a mismatch in the slopes at every single one of these 5 points. Hence, $f(x)$ is non-differentiable at exactly 5 points. Thus, (B) is TRUE.
- Global Maximum: The function increases up to $x=\phi-1$, where it attains its maximum value $f(\phi-1)=\phi=\dfrac{\sqrt{5}+1}{2}$. Statement (C) claims the maximum value is $\dfrac{\sqrt{5}-1}{2}$, making it FALSE.
- Monotonicity: For $x>1$, $f(x)=\dfrac{2}{1+x} \implies f'(x)=-\dfrac{2}{(1+x)^2}<0$, meaning it is strictly decreasing. Thus, (D) is TRUE.

Q7. Let A, B, and C be three mutually independent events with probabilities $P(A)=\dfrac{1}{2}$, $P(B)=\dfrac{1}{3}$, and $P(C)=\dfrac{1}{4}$. Define the following events:

- E_1: Both A and B occur, or neither of them occurs.
- E_2: Exactly one of the events B or C occurs.
- E_3: At least one of the events A or C occurs.

Then which of the following statements is (are) **TRUE**?

(A) E_1 and B are independent events. (B) E_1 and E_2 are independent events.

(C) $P(E_3|E_1)=P(E_3)$ (D) $P(E_2|E_1^c)=\dfrac{5}{12}$

Ans. A, B, D

Sol. From the given probabilities, we have $P(A^c)=\dfrac{1}{2}$, $P(B^c)=\dfrac{2}{3}$, and $P(C^c)=\dfrac{3}{4}$.

For E_1:

$$E_1 = (A \cap B) \cup (A^c \cap B^c)$$

$$\begin{aligned} P(E_1) &= P(A)P(B) + P(A^c)P(B^c) \\ &= \frac{1}{2} \times \frac{1}{3} + \frac{1}{2} \times \frac{2}{3} \\ &= \frac{1}{2} \end{aligned}$$

For statement (A):

$$E_1 \cap B = A \cap B \implies P(E_1 \cap B) = P(A)P(B) = \frac{1}{6}$$

Since $P(E_1)P(B) = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6} = P(E_1 \cap B)$, E_1 and B are independent. Thus, (A) is true.

For statement (B):

$$\begin{aligned} E_2 &= (B \cap C^c) \cup (B^c \cap C) \\ &\implies P(E_2) \\ &= \frac{1}{3} \times \frac{3}{4} + \frac{2}{3} \times \frac{1}{4} \\ &= \frac{5}{12} \end{aligned}$$

$$E_1 \cap E_2 = (A \cap B \cap C^c) \cup (A^c \cap B^c \cap C)$$

$$\begin{aligned} P(E_1 \cap E_2) &= \frac{1}{2} \times \frac{1}{3} \times \frac{3}{4} + \frac{1}{2} \times \frac{2}{3} \times \frac{1}{4} \\ &= \frac{3}{24} + \frac{2}{24} \\ &= \frac{5}{24} \end{aligned}$$

Since $P(E_1)P(E_2) = \frac{1}{2} \times \frac{5}{12} = \frac{5}{24} = P(E_1 \cap E_2)$, E_1 and E_2 are independent. Thus, (B) is true.

For statement (C):

$$\begin{aligned} E_3 &= A \cup C \\ &\implies P(E_3) \\ &= \frac{1}{2} + \frac{1}{4} - \frac{1}{8} \\ &= \frac{5}{8} \end{aligned}$$

$$E_1 \cap E_3 = (A \cap B) \cup (A^c \cap B^c \cap C)$$

$$\begin{aligned} P(E_1 \cap E_3) &= \frac{1}{6} + \frac{1}{2} \times \frac{2}{3} \times \frac{1}{4} \\ &= \frac{1}{6} + \frac{1}{12} \\ &= \frac{1}{4} \end{aligned}$$

$$\begin{aligned} P(E_3|E_1) &= \frac{P(E_1 \cap E_3)}{P(E_1)} \\ &= \frac{1/4}{1/2} \\ &= \frac{1}{2} \\ &\neq P(E_3) \end{aligned}$$

Thus, (C) is false.

For statement (D):

Since E_1 and E_2 are independent, E_1^c and E_2 are also independent.

$P(E_2|E_1^c) = P(E_2) = \dfrac{5}{12}$

Thus, (D) is true.

Q8. Let X be a discrete random variable taking values in the set of positive integers $\{1, 2, 3, \ldots\}$ such that its probability mass function is given by $P(X = k) = \dfrac{4k}{3^{k+1}}, \quad k \in \{1, 2, 3, \ldots\}$ Which of the following statements is (are) **TRUE**?

(A) The mean of X is 2 and the variance of X is $\dfrac{3}{2}$.

(B) The probability that X takes an even value is $\dfrac{3}{8}$.

(C) $\displaystyle\sum_{k=1}^{\infty} P(X > k) = 1.$

(D) The probability that X takes an even value is $\dfrac{5}{8}$.

Ans. A, B, C

Sol. Using the standard arithmetico-geometric series expansion for $|x| < 1$:

$\displaystyle\sum_{k=1}^{\infty} kx^k = \frac{x}{(1-x)^2} \quad \text{and} \quad \sum_{k=1}^{\infty} k^2 x^k = \frac{x(1+x)}{(1-x)^3}$

For Statement (A):

The mean $E[X]$ is given by:

$E[X] = \displaystyle\sum_{k=1}^{\infty} k \cdot \frac{4k}{3^{k+1}} = \frac{4}{3} \sum_{k=1}^{\infty} k^2 \left(\frac{1}{3}\right)^k = \frac{4}{3} \cdot \frac{\frac{1}{3}\left(1 + \frac{1}{3}\right)}{\left(1 - \frac{1}{3}\right)^3} = \frac{4}{3} \cdot \frac{\frac{4}{9}}{\frac{8}{27}} = 2$

To find the variance, we first compute $E[X^2]$ using $\displaystyle\sum_{k=1}^{\infty} k^3 x^k = \frac{x(1 + 4x + x^2)}{(1-x)^4}$:

$E[X^2] = \dfrac{4}{3} \displaystyle\sum_{k=1}^{\infty} k^3 \left(\frac{1}{3}\right)^k = \frac{4}{3} \cdot \frac{\frac{1}{3}\left(1 + \frac{4}{3} + \frac{1}{9}\right)}{\left(1 - \frac{1}{3}\right)^4} = \frac{4}{3} \cdot \frac{\frac{22}{27}}{\frac{16}{81}} = \frac{11}{2}$

$\mathrm{Var}(X) = E[X^2] - (E[X])^2 = \dfrac{11}{2} - 2^2 = \dfrac{3}{2}$

Thus, statement (A) is correct.

For Statement (B):

$P(X \text{ is even}) = \displaystyle\sum_{m=1}^{\infty} P(X = 2m) = \sum_{m=1}^{\infty} \frac{4(2m)}{3^{2m+1}} = \frac{8}{3} \sum_{m=1}^{\infty} m \left(\frac{1}{9}\right)^m = \frac{8}{3} \cdot \frac{\frac{1}{9}}{\left(1 - \frac{1}{9}\right)^2} = \frac{8}{3} \cdot \frac{9}{64} = \frac{3}{8}$

Thus, statement (B) is correct. For Statement (C):

For any random variable X taking values in $\{1, 2, 3, \ldots\}$, we have the identity:

$E[X] = \displaystyle\sum_{k=0}^{\infty} P(X > k) = P(X > 0) + \sum_{k=1}^{\infty} P(X > k)$

Since $X \geq 1$ with probability 1, $P(X > 0) = 1$. Substituting $E[X] = 2$:

$2 = 1 + \displaystyle\sum_{k=1}^{\infty} P(X > k) \implies \sum_{k=1}^{\infty} P(X > k) = 1$

Thus, statement (C) is correct.

Paragraph

Consider the functions f(x) and g(x) defined for all $x \geq 0$ as follows:

$f(x) = \max_{0 \leq t \leq x} \{3t^2 - t^3\}$

$g(x) = \min_{t \in [x, x+2]} \{2t^3 - 9t^2 + 12t\}$

Based on the above information, answer the following two questions:

Q9. The value of $\lim\limits_{x \to 2^+} \dfrac{g(x) - g(2)}{(x-2)^2}$ is :

(A) 0 (B) 1 (C) 3 (D) 5

Ans. C

Sol. Let $h(t) = 3t^2 - t^3 \quad \Rightarrow h'(t) = 3t(2 - t)$.

For $t \in [0, 2]$, $h'(t) \geq 0$, which means h(t) is strictly increasing. Thus, for $x \in [0, 2]$, the maximum value in [0, x] occurs at the right endpoint : $f(x) = 3x^2 - x^3$

Let $k(t) = 2t^3 - 9t^2 + 12t \quad \Rightarrow k'(t) = 6(t - 1)(t - 2)$.

The function k(t) strictly increases on [0, 1] and [2, ∞) and strictly decreases on [1, 2]. Note that k(1) = 5,

k(2) = 4, and k(1/2) = 4.

Solution for Question 1:

For $x \geq 2$, the interval [x, x + 2] lies entirely within [2, ∞) where k(t) is strictly increasing. Thus, the minimum occurs at the left endpoint :

$g(x) = k(x) = 2x^3 - 9x^2 + 12x \quad \Rightarrow g(2) = 4$

$$\lim_{x \to 2^+} \frac{g(x) - g(2)}{(x-2)^2} = \lim_{x \to 2^+} \frac{2x^3 - 9x^2 + 12x - 4}{(x-2)^2}$$

$$= \lim_{x \to 2^+} \frac{(2x-1)(x-2)^2}{(x-2)^2} = \lim_{x \to 2^+} (2x - 1) = 3$$

Q10. The greatest value of the function $F(x) = g(x) - f(x)$ on the interval [0, 2] is :

(A) 4 (B) $\dfrac{27}{8}$ (C) $\dfrac{15}{4}$ (D) 3

Ans. B

Sol. Now analyze $g(x) = \min_{t \in [x, x+2]} k(t)$ for $x \in [0, 2]$:

If $x \in [0, 1/2]$, then $x + 2 \geq 2$. The interval contains t = 2 (local minimum, k(2) = 4). Since $k(x) \leq k(1/2) = 4$, the absolute minimum on [x, x + 2] is achieved at t = x. Hence, $g(x) = 2x^3 - 9x^2 + 12x$.

If $x \in (1/2, 2]$, then $k(x) > 4$ and $2 \in [x, x + 2]$. The absolute minimum value on this interval is k(2) = 4. Hence,

g(x) = 4.

Forming the difference function $F(x) = g(x) - f(x)$ on [0, 2]:

For $x \in [0, 1/2]$: $F(x) = (2x^3 - 9x^2 + 12x) - (3x^2 - x^3) = 3x^3 - 12x^2 + 12x$

$F'(x) = 9x^2 - 24x + 12 = 3(3x - 2)(x - 2)$

Since $F'(x) > 0$ for $x \in [0, 1/2]$, $F(x)$ strictly increases to $F\left(\frac{1}{2}\right) = \frac{27}{8}$.

For $x \in [1/2, 2]$: $F(x) = 4 - (3x^2 - x^3) = x^3 - 3x^2 + 4 \Rightarrow$ $F'(x) = 3x(x - 2)$

Since $F'(x) < 0$ for $x \in (1/2, 2)$, $F(x)$ strictly decreases from $F(1/2) = \frac{27}{8}$ to $F(2) = 0$.

Thus, the greatest value of $F(x)$ on $[0, 2]$ is $\frac{27}{8}$.

Q1. Let $I(x) = \int x^5 e^{x^3} dx$ be an indefinite integral satisfying $I(0) = 0$. If $\lim\limits_{x \to 0} \dfrac{18I(x) - 3x^6}{x^9} = L$, then the value of L is

Ans. 2

Sol. To evaluate the integral $I(x) = \int x^3 \cdot e^{x^3} \cdot x^2 dx$, we use the substitution $x^3 = t$, which gives $3x^2 dx = dt$:

$$I(x) = \frac{1}{3} \int te^t dt$$

$$I(x) = \frac{1}{3}(t - 1)e^t + C = \frac{1}{3}(x^3 - 1)e^{x^3} + C$$

Applying the boundary condition $I(0) = 0$:

$$0 = \frac{1}{3}(0 - 1)e^0 + C \implies C = \frac{1}{3}$$

Thus, the function is:

$$I(x) = \frac{1}{3}(x^3 - 1)e^{x^3} + \frac{1}{3}$$

$$18I(x) = 6(x^3 - 1)e^{x^3} + 6$$

Using the Taylor series expansion for $e^{x^3} = 1 + x^3 + \dfrac{x^6}{2} + \dfrac{x^9}{6} + \ldots$:

$$\begin{aligned} L &= \lim_{x \to 0} \frac{(3x^6 + 2x^9 + \ldots) - 3x^6}{x^9} \\ &= \lim_{x \to 0} \frac{2x^9 + \ldots}{x^9} \\ &= 2 \end{aligned}$$

Q2. Let L_1 and L_2 be two parallel lines given by $x - y + 2 = 0$ and $x - y - 4 = 0$ respectively, and let L_3 be the line $x + y = 0$. A variable point $P(x, y)$ moves in the region contained between the parallel lines L_1 and L_2 such that the square of its perpendicular distance from L_3 is equal to the product of its perpendicular distances from L_1 and L_2. If the locus of P is a circle of radius r, then the value of $2r^2$ is equal to _____.

Ans. 9

Sol. The perpendicular distances of the point $P(x, y)$ from the lines L_1, L_2, and L_3 are given by:

$$d_1 = \frac{|x-y+2|}{\sqrt{2}}, \quad d_2 = \frac{|x-y-4|}{\sqrt{2}}, \quad d_3 = \frac{|x+y|}{\sqrt{2}}$$

Since the point $P(x, y)$ lies in the region between the parallel lines L_1 and L_2, the expressions $(x-y+2)$ and $(x-y-4)$ must have opposite signs. Therefore, their product is negative:

$$|(x-y+2)(x-y-4)| = -(x-y+2)(x-y-4)$$

According to the given condition, $d_3^2 = d_1 \cdot d_2$:

$$\left(\frac{x+y}{\sqrt{2}}\right)^2 = \frac{-(x-y+2)(x-y-4)}{2} (x+y)^2 = -[(x-y)^2 - 2(x-y) - 8] \quad \Rightarrow$$

$$(x+y)^2 + (x-y)^2 - 2(x-y) - 8 = 0$$

$$\Rightarrow x^2 + y^2 - x + y - 4 = 0$$

This represents a circle with radius r given by:

$$r = \sqrt{\left(-\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 - (-4)} = \sqrt{\frac{1}{4} + \frac{1}{4} + 4} = \sqrt{\frac{9}{2}}$$

Thus, the value of $2r^2$ is:

$$2r^2 = 2 \times \frac{9}{2} = 9$$

Q3. Consider a continuous frequency distribution consisting of four classes $[0, 10)$, $[10, 20)$, $[20, 30)$, and $[30, 40)$ with frequencies f_1, f_2, f_3, and f_4 respectively, where:

- f_1 is the number of points of non-differentiability of the function $g(x) = |x^3 - 3x^2 + 2x|$ in $\mathbb{R}$.
- f_2 is the number of integers in the domain of the function $h(x) = \sqrt{\frac{12 + 4x - x^2}{\log_e x}}$.
- f_3 is the value of the limit $\lim\limits_{x \to 0} \frac{\sqrt{1+x^2} - 1}{1 - \cos x}$.
- f_4 is the number of distinct real roots of the equation $e^x + x^2 - x - 1 = 0$.

If the median of this distribution is M, then the value of $(M - 10)$ is equal to _____.

Ans. 4

Sol. The function inside the absolute value is $x^3 - 3x^2 + 2x = x(x-1)(x-2)$. Since it has three distinct simple real roots at $x = 0, 1, 2$, the function $g(x) = |x(x-1)(x-2)|$ is non-differentiable at exactly these 3 points.

Thus, $f_1 = 3$.

For $h(x) = \sqrt{\frac{12 + 4x - x^2}{\log_e x}}$ to be defined, we require $x > 0$ and $x \neq 1$.

- Case I: If $x > 1$, then $\log_e x > 0$. We must have
  $12 + 4x - x^2 \geq 0 \implies x^2 - 4x - 12 \leq 0 \implies (x-6)(x+2) \leq 0 \implies x \in [-2, 6]$. Taking intersection with $x > 1$ gives $x \in (1, 6]$. The integers in this interval are $\{2, 3, 4, 5, 6\}$.
- Case II: If $0 < x < 1$, then $\log_e x < 0$. We need $12 + 4x - x^2 \leq 0 \implies x \in (-\infty, -2] \cup [6, \infty)$, which yields no solution in $(0, 1)$.

  Thus, the domain contains exactly 5 integers, so $f_2 = 5$.

  Using Taylor series expansions as $x \to 0$:
  $\lim\limits_{x \to 0} \dfrac{\sqrt{1+x^2} - 1}{1 - \cos x} = \lim\limits_{x \to 0} \dfrac{(1 + \frac{1}{2}x^2 + \ldots) - 1}{\frac{1}{2}x^2 + \ldots} = 1$

  Thus, $f_3 = 1$.

  Let $k(x) = e^x + x^2 - x - 1$. Note that $k(0) = 0$ and $k'(x) = e^x + 2x - 1 \implies k'(0) = 0$.

  Since $k''(x) = e^x + 2 > 0$ for all $x \in \mathbb{R}$, $k(x)$ is strictly convex. Therefore, $x = 0$ is the unique global minimum, meaning $k(x) > 0$ for all $x \neq 0$. Thus, $x = 0$ is the only real root.

  Hence, $f_4 = 1$.

  The frequency table is given below:

Class	**Frequency (f)**	**Cumulative Frequency (CF)**
$[0, 10)$	3	3
$[10, 20)$	5	8
$[20, 30)$	1	9
$[30, 40)$	1	10

  Total frequency $N = 10 \implies \dfrac{N}{2} = 5$.

  The median class is $[10, 20)$ since its cumulative frequency is the first to exceed 5.

  Using the median formula:
  $M = L + \left(\dfrac{\frac{N}{2} - CF}{f} \right) \times h$
  $M = 10 + \left(\dfrac{5 - 3}{5} \right) \times 10 = 10 + 4 = 14$

  Thus, $M - 10 = 14 - 10 = 4$.

Q4. Let $f(x) = (x-1)(x-2)(x-3)(x-4)(x-5)$. Find the number of distinct real roots of the equation $(f'(x))^2 + f(x)f''(x) = 0$

Ans. 8

Sol. Let us define an auxiliary function $g(x) = f(x)f'(x)$.

Differentiating $g(x)$ with respect to x using the product rule, we get:

$g'(x) = f'(x)f'(x) + f(x)f''(x) = (f'(x))^2 + f(x)f''(x)$

Thus, the given equation is equivalent to finding the number of distinct real roots of $g'(x) = 0$.

1. The polynomial $f(x)$ is of degree 5 and has 5 distinct real roots: $1, 2, 3, 4, 5$.

2. By Rolle's Theorem, since $f(x)$ is continuous and differentiable everywhere, its derivative $f'(x) = 0$ must have at least one real root in each of the intervals $(1, 2)$, $(2, 3)$, $(3, 4)$, and $(4, 5)$. Since $f'(x)$ is of degree 4, it has exactly 4 distinct real roots. Let these roots be r_1, r_2, r_3, r_4 such that:
   $1 < r_1 < 2 < r_2 < 3 < r_3 < 4 < r_4 < 5$
3. The roots of $g(x) = 0$ are the combined roots of $f(x) = 0$ and $f'(x) = 0$. Because the roots interlace, all $5 + 4 = 9$ roots of $g(x)$ are real and distinct. Ordered chronologically, they are:
   $1, r_1, 2, r_2, 3, r_3, 4, r_4, 5$
4. By Rolle's Theorem, between any two consecutive distinct roots of $g(x) = 0$, there exists at least one real root of $g'(x) = 0$. Since there are 9 distinct roots, they form 8 disjoint intervals. Therefore, $g'(x) = 0$ must have at least 8 distinct real roots.
5. Since $g(x)$ is a polynomial of degree 9, its derivative $g'(x)$ is a polynomial of degree 8. A polynomial of degree 8 can have at most 8 roots.
   Consequently, $g'(x) = 0$ has exactly 8 distinct real roots.

Q1. Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined by $f(x) = \begin{cases} x^3 + x & \text{if } x \geq 0 \\ 2x & \text{if } x < 0 \end{cases}$ Let $g : \mathbb{R} \to \mathbb{R}$ be the inverse function of $f(x)$, and let $h : \mathbb{R} \to \mathbb{R}$ be a composite function defined by $h(x) = g(x^3 - 3x)$. If S is the set of all points in $\mathbb{R}$ where $h(x)$ is not differentiable, then the value of $\sum_{x \in S} x^2$ is equal to _____.

Ans. 6

Sol. Since $f'(x) = 3x^2 + 1 > 0$ for $x > 0$ and $f'(x) = 2 > 0$ for $x < 0$, the function $f(x)$ is strictly increasing and continuous on $\mathbb{R}$. Thus, it is a bijection, and its inverse function $g(x)$ exists for all $x \in \mathbb{R}$.

To find the points where $g(x)$ is not differentiable, we examine the one-sided derivatives at $x = 0$ (since $f(0) = 0$):

$g'(0^+) = \dfrac{1}{f'(0^+)} = \dfrac{1}{1} = 1$

$g'(0^-) = \dfrac{1}{f'(0^-)} = \dfrac{1}{2}$

Since $g'(0^+) \neq g'(0^-)$, the inverse function $g(x)$ is not differentiable only at $x = 0$. For all other points $x \neq 0$, $g(x)$ is differentiable.

Now, consider the composite function $h(x) = g(\phi(x))$, where $\phi(x) = x^3 - 3x$. Since $\phi(x)$ is a polynomial, it is differentiable everywhere. By the chain rule, $h(x)$ can only be non-differentiable at points where $\phi(x) = 0$:

$x^3 - 3x = 0 \implies x(x^2 - 3) = 0 \implies x = 0, \sqrt{3}, -\sqrt{3}$

Let's check the differentiability of $h(x)$ at these three critical points by looking at the behavior of $\phi(x)$ around them:

1. At $x = 0$:

$\phi(0) = 0$ and $\phi'(0) = -3$. Since $\phi'(0) < 0$, for a small $x > 0$, $\phi(x) < 0$, and for a small $x < 0$, $\phi(x) > 0$.

$h'(0^+) = g'(0^-) \cdot \phi'(0) = \dfrac{1}{2}(-3) = -1.5$

$h'(0^-) = g'(0^+) \cdot \phi'(0) = 1(-3) = -3$

Since $h'(0^+) \neq h'(0^-)$, $h(x)$ is not differentiable at $x = 0$.

2. At $x = \sqrt{3}$:

$\phi(\sqrt{3}) = 0$ and $\phi'(\sqrt{3}) = 6 > 0$. For $x > \sqrt{3}$, $\phi(x) > 0$, and for $x < \sqrt{3}$, $\phi(x) < 0$.

$h'(\sqrt{3}^+) = g'(0^+) \cdot \phi'(\sqrt{3}) = 1(6) = 6$

$h'(\sqrt{3}^-) = g'(0^-) \cdot \phi'(\sqrt{3}) = \frac{1}{2}(6) = 3$

Since $h'(\sqrt{3}^+) \neq h'(\sqrt{3}^-)$, $h(x)$ is not differentiable at $x = \sqrt{3}$.

3. At $x = -\sqrt{3}$:

$\phi(-\sqrt{3}) = 0$ and $\phi'(-\sqrt{3}) = 6 > 0$. Similarly, the left and right derivatives will not match (3 and 6), so $h(x)$ is not differentiable at $x = -\sqrt{3}$.

Thus, the set of non-differentiable points is $S = \{0, \sqrt{3}, -\sqrt{3}\}$.

The required value is:

$\sum_{x \in S} x^2 = 0^2 + (\sqrt{3})^2 + (-\sqrt{3})^2 = 0 + 3 + 3 = 6$

Q2. Consider the parabola $C_1 : y = x^2$ and a variable circle $C_2 : (x-k)^2 + \left(y - \frac{7}{2}\right)^2 = 1$, where $k \in \mathbb{R}$. Let $f(x)$ denote the square of the distance between the center of the circle C_2 and a variable point (x, x^2) on the parabola C_1. Let n be the number of distinct points of local minima of the function $f(x)$. If S is the set of all integral values of k for which $n = 2$, then the sum of k^2 for all $k \in S$ is equal to _____.

Ans. 28

Sol. The center of the circle C_2 is $P\left(k, \frac{7}{2}\right)$. The square of the distance from P to any point (x, x^2) on the parabola C_1 is given by:

$f(x) = (x-k)^2 + \left(x^2 - \frac{7}{2}\right)^2$

$f'(x) = 2(x-k) + 2\left(x^2 - \frac{7}{2}\right)(2x) = 4x^3 - 12x - 2k$

For critical points, setting $f'(x) = 0$

$2x^3 - 6x - k = 0 \implies 2x^3 - 6x = k$

Let $g(x) = 2x^3 - 6x$. Then $g'(x) = 6x^2 - 6 = 0 \implies x = \pm 1$.

The local maximum value is $g(-1) = 4$ and the local minimum value is $g(1) = -4$.

To find the nature of these critical points, we compute the second derivative of $f(x)$:

$f''(x) = 12x^2 - 12 = 12(x^2 - 1)$

• Case I: If $k \in (-4, 4)$, the cubic equation $g(x) = k$ has three distinct real roots $x_1 < x_2 < x_3$ such that $x_1 \in (-2, -1)$, $x_2 \in (-1, 1)$, and $x_3 \in (1, 2)$.

For x_1: $x_1 < -1 \implies x_1^2 > 1 \implies f''(x_1) > 0$ (Local Minimum).

For x_2: $-1 < x_2 < 1 \implies x_2^2 < 1 \implies f''(x_2) < 0$ (Local Maximum).

For x_3: $x_3 > 1 \implies x_3^2 > 1 \implies f''(x_3) > 0$ (Local Minimum).

Thus, $n = 2$ distinct points of local minima exist when $k \in (-4, 4)$.

• Case II: If $|k| \geq 4$, the equation has only one real root where $f''(x) \geq 0$, giving $n = 1$.

Hence, $n = 2$ if and only if $k \in (-4, 4)$.

The integral values of k in this interval are $S = \{-3, -2, -1, 0, 1, 2, 3\}$.

The required sum of k^2 is:

$$\sum_{k \in S} k^2 = (-3)^2 + (-2)^2 + (-1)^2 + 0^2 + 1^2 + 2^2 + 3^2 = 9 + 4 + 1 + 0 + 1 + 4 + 9 = 28$$

Q3. Let $A(3, 4)$ be a fixed point in the xy-plane. Let B be a variable point on the line $y = x$ and C be a variable point on the line $y = -x$ such that $\angle BAC = 90^\circ$. If the locus of the midpoint of the line segment BC is a straight line L, then the perpendicular distance of the point $A(3, 4)$ from the line L is equal to _____.

Ans. 2.5

Sol. Let the coordinates of the variable points B and C be $B(t, t)$ and $C(s, -s)$ respectively, where $t, s \in \mathbb{R}$.

The vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are given by:

$$\overrightarrow{AB} = (t-3)\hat{i} + (t-4)\hat{j} \overrightarrow{AC} = (s-3)\hat{i} + (-s-4)\hat{j}$$

Since $\angle BAC = 90^\circ$, $\overrightarrow{AB} \cdot \overrightarrow{AC} = 0 \Rightarrow (t-3)(s-3) + (t-4)(-s-4) = 0$

$\Rightarrow ts - 3t - 3s + 9 - ts - 4t + 4s + 16 = 0 \Rightarrow -7t + s + 25 = 0 \implies s = 7t - 25$

Let $M(x, y)$ be the midpoint of the segment BC. Therefore:

$$x = \frac{t+s}{2} = \frac{t + (7t - 25)}{2} = 4t - \frac{25}{2} y = \frac{t-s}{2} = \frac{t - (7t - 25)}{2} = -3t + \frac{25}{2}$$

eliminate the parameter t,

$\Rightarrow L \equiv 6x + 8y - 25 = 0$

The perpendicular distance p from the point $A(3, 4)$ to the line L is:

$$p = \frac{|6(3) + 8(4) - 25|}{\sqrt{6^2 + 8^2}} p = \frac{|18 + 32 - 25|}{10} = \frac{25}{10} = 2.5$$